FACULTY OF COMPUTER APPLICATIONS

Marwadi University
Marwadi Chandarana Group
NAAC A+

Faculty of Computer Applications

MCA Sem-1
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Lab Book

Sub Code: 05MC0108
(DBMS)

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Index

Unit : 1 Exercise No	Unit : 1 Definition
1	Write a PL/SQL block that find 1. Area Of Circle 2. Area Of Rectangle 3. Circumference of Circle
2	Write a PL/SQL block to accept product name, qty and price from user and then calculate discount in Rs. based on the given (%).
3	Write a PL/SQL block to calculate the total, percentage and grade of student based on his/her Rollno from RESULT table. (Create RESULT table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per, Grade attributes with appropriate data type).
4	1. Write a program that prints value 1 to 100 numbers using FOR LOOP. 2. Write a program that prints value 1 to 100 number using LOOP Command. 3. Write a program that prints value 1 to 100 number using WHILE LOOP Command
5	Write a PL/SQL block which displays gross salary of employees as per user input EID. (Consider EMP table with EID, EName, Deptno,Deptname Gender, Age, BasicSal) with appropriate data types.) Gross_Salary: BASICSAL + (DA + HRA + Medical) – PF. Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.
6	Write a PL/SQL block which accepts measurement in feet and displays it in cm, inch and meter.
7	Write a PL/SQL block which converts temperature from Celsius to Fahrenheit.
8	Write a PL/SQL block to delete the record of employee for given EID.
9	Write a PL/SQL block that calculates the simple interest based on given principal amount, rate of interest and number of years.
10	Write a PL/SQL block to calculate the square and cube of the given number.

11	Write a simple procedure that increases by the salary of employees for the given department no. by percentage inputted by the user using IN parameter. Also handle the exception if inputted department number not found.
12	Write a procedure that search whether the given employee number is present or not in the table if present then print name, salary and department number of that employee. (Use both IN and OUT mode variables) and also Write a PL/SQL block to call the SEARCH_EMP procedure.
13	Write a procedure that accept employee ID and remove records from employee table. Also create a calling program to call procedure.
14	Write a function that returns the age, category of an employee. Categories: ● Age < 18 → Minor ● 18–60 → Adult ● 60 → Senior
15	Write a PACKAGE which includes Package Declaration, Package Body and call the individual object from the Package.

1. Write a PL/SQL block that find:
1. Area of Circle
2. Area of Rectangle
3. Circumference of Circle

```
/*
	Write a PL/SQL block that find
	1.Area of circle(pi*radius*radius)
	2.Area of Rectangle(length*width)
	3.Circumference of circle(2*PI*RADIUS)
*/

ACCEPT CHOICE PROMPT "Enter Your Choice: 1.Area of circle ,2.Area of Rectangle
,3.Circumference of circle "
SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

DECLARE
	CHOICE NUMBER:=&CHOICE;
	PI NUMBER(7,2):=3.14;
	RADIUS NUMBER(5);
	LENGTH NUMBER(5);
	WIDTH NUMBER(5);
BEGIN
	--CHOICE:=&CHOICE;

	DBMS_OUTPUT.PUT_LINE('    ');
	IF (CHOICE=1) THEN
		RADIUS:=&RADIUS;
		DBMS_OUTPUT.PUT_LINE('Area of circle is : ' ||PI*(POWER(RADIUS,2)));
	ELSIF (CHOICE=2) THEN
		LENGTH:=&LENGTH;
		WIDTH:=&WIDTH;
		DBMS_OUTPUT.PUT_LINE('Area of Rectangle is : ' ||LENGTH*WIDTH);
	ELSIF (CHOICE=3) THEN
		DBMS_OUTPUT.PUT_LINE('Circumference of circle is : ' ||2*PI*RADIUS);
	ELSE
		DBMS_OUTPUT.PUT_LINE('Invalid Choice');
```

```
        END IF;
END;
/
SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p1.sql
Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :-  1
Enter value for radius: 12
Enter value for length: 8
Enter value for width: 5

Area of circle is : 452.16
```

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p1.sql
Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :-  2
Enter value for radius: 12
Enter value for length: 45
Enter value for width: 40

Area of Rectangle is : 1800
```

```
SQL>  @ D:\Kunjal\DBMS\Unit_1\p1.sql

Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :-  3
Enter value for radius: 10
Enter value for length: 40
Enter value for width: 45

Circumference of circle is : 62.8
```

2. Write a PL/SQL block to accept product name, qty and price from user and then calculate discount in Rs. based on the given (%).

```
/*
  Write a PL/SQL  block to accept product name ,qty and price from user
   and then calculate discount in Rs.based on the given (%)
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

ACCEPT PN PROMPT "Enter product name :- "
DECLARE
        P_NAME          VARCHAR(20);
        QTY     NUMBER(5);
        PRICE   NUMBER(7,2);
        TP      NUMBER(7,2);
        DISCOUNT        NUMBER(5);
        FP      NUMBER(7,2);
BEGIN
        P_NAME:='&PN';
        QTY:=&QTY;
        PRICE:=&PRICE;
        DISCOUNT:=&DISCOUNT;

        TP:=PRICE*QTY;
        FP:=TP-(TP*DISCOUNT/100);
        DBMS_OUTPUT.PUT_LINE('    ');
        DBMS_OUTPUT.PUT_LINE('Product name is := '||P_NAME);
        DBMS_OUTPUT.PUT_LINE('Total qty is := '||QTY);
        DBMS_OUTPUT.PUT_LINE('Price is := '||PRICE);
        DBMS_OUTPUT.PUT_LINE('Total Price is := '||TP);
        DBMS_OUTPUT.PUT_LINE('Discount is := '||DISCOUNT);
        DBMS_OUTPUT.PUT_LINE('Final Price is := '||FP);

END;
/
SET SERVEROUTPUT OFF
```

SET VERIFY ON
SET FEEDBACK ON

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p2.sql
Enter product name :- key-board
Enter value for qty: 3
Enter value for price: 200
Enter value for discount: 12

Product name is := key-board
Total qty is := 3
Price is := 200
Total Price is := 600
Discount is := 12
Final Price is := 528
```

3. Write a PL/SQL block to calculate the total, percentage and grade of student based on his/her Rollno from RESULT table. (Create RESULT table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per, Grade attributes with appropriate data type).

```
/*
 Write a PL/SQL block to calculate the total, percentage and grade of
student based on his/her Rollno from RESULT table. (Create RESULT
table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per,
Grade attributes with appropriate data type).
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF
DECLARE
         ROLL_NO NUMBER(5);
         NAME VARCHAR2(30);
         SUB1 NUMBER(3);
         SUB2 NUMBER(3);
          SUB3 NUMBER(3);
         SUB4 NUMBER(3);
         SUB5 NUMBER(3);
        TOTAL NUMBER(5);
        PER NUMBER(7,2);
       GRADE VARCHAR2(30);
BEGIN
       ROLL_NO:=&ROLL_NO;
       NAME:='&NAME';
       SUB1:=&SUB1;
       SUB2 :=&SUB2 ;
       SUB3 :=&SUB3 ;
       SUB4:=&SUB4;
       SUB5:=&SUB5;
       TOTAL:=SUB1+SUB2+SUB3+SUB4+SUB5;
       PER:=TOTAL/5;

       IF (PER>=75) THEN
               GRADE:='O';
       ELSIF (PER<=75)AND(PER>=70) THEN
               GRADE:='A';
```

```
        ELSIF (PER<=60)AND(PER>=70) THEN
                GRADE:='B';
        ELSIF (PER<=50)AND(PER>=60) THEN
                GRADE:='C';
        ELSE
                GRADE:='PASS CLASS';
        END IF;

        INSERT INTO
RESULT(ROLLNO,NAME,SUB1,SUB2,SUB3,SUB4,SUB5,TOTAL,PER,GRADE)VALUES(RO
LL_NO,NAME,SUB1,SUB2,SUB3,SUB4,SUB5,TOTAL,PER,GRADE);

        DBMS_OUTPUT.PUT_LINE('    ');
        DBMS_OUTPUT.PUT_LINE('Roll no:-'||ROLL_NO);
        DBMS_OUTPUT.PUT_LINE('Name:-'||NAME);
        DBMS_OUTPUT.PUT_LINE('    ');
        DBMS_OUTPUT.PUT_LINE('---------------------Result--------------------------');
        DBMS_OUTPUT.PUT_LINE('Marks of sub1:-'||SUB1);
        DBMS_OUTPUT.PUT_LINE('Marks of sub2:-'||SUB2);
        DBMS_OUTPUT.PUT_LINE('Marks of sub3:-'||SUB3);
        DBMS_OUTPUT.PUT_LINE('Marks of sub4:-'||SUB4);
        DBMS_OUTPUT.PUT_LINE('Marks of sub5:-'||SUB5);

        DBMS_OUTPUT.PUT_LINE('    ');
        DBMS_OUTPUT.PUT_LINE('------------------------------------------------');
        DBMS_OUTPUT.PUT_LINE('Your total marks : '||TOTAL);
        DBMS_OUTPUT.PUT_LINE('Percentage :'||PER||'%');
        DBMS_OUTPUT.PUT_LINE('Grade:'||GRADE);
END;
/

SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p3.sql
Enter value for roll_no: 17
Enter value for name: kunjal
Enter value for sub1: 80
Enter value for sub2: 90
Enter value for sub3: 88
Enter value for sub4: 96
Enter value for sub5: 95

Roll no:-17
Name:-kunjal

---------------------Result--------------------------
Marks of sub1:-80
Marks of sub2:-90
Marks of sub3:-88
Marks of sub4:-96
Marks of sub5:-95

-----------------------------------------------
Your total marks : 449
Percentage :89.8%
Grade:O
```

4. 1. Write a program that prints value 1 to 100 numbers using FOR LOOP.
2. Write a program that prints value 1 to 100 number using LOOP Command.
3. Write a program that prints value 1 to 100 number using WHILE LOOP Command

---

```
/*
1. Write a program that prints value 1 to 100 numbers using FOR LOOP.
2. Write a program that prints value 1 to 100 number using LOOP
Command.
3. Write a program that prints value 1 to 100 number using WHILE LOOP
Command
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

DECLARE
   i NUMBER := 1;
BEGIN

   -- FOR LOOP
   DBMS_OUTPUT.PUT_LINE('--- FOR LOOP ---');
   FOR i IN 1..100 LOOP
      DBMS_OUTPUT.PUT(i || ' ');
   END LOOP;
   DBMS_OUTPUT.PUT_LINE('                        ');


   -- SIMPLE LOOP
   DBMS_OUTPUT.PUT_LINE('--- SIMPLE LOOP ---');
   i := 1;
   LOOP
      DBMS_OUTPUT.PUT(i || ' ');
      i := i + 1;
      EXIT WHEN i > 100;
   END LOOP;
   DBMS_OUTPUT.PUT_LINE('');
```

```
 DBMS_OUTPUT.PUT_LINE('--- WHILE LOOP ---');
  i := 1;
WHILE i <= 100 LOOP
        DBMS_OUTPUT.PUT(i || ' ');
        i := i + 1;
    END LOOP;
      DBMS_OUTPUT.PUT_LINE('            ');
END;
/
SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p4.sql
--- FOR LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
--- SIMPLE LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
--- WHILE LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
```

5. Write a PL/SQL block which displays gross salary of employees as per user input EID. (Consider EMP table with EID, EName, Deptno,Deptname Gender, Age, BasicSal) with appropriate data types.) Gross_Salary: BASICSAL + (DA + HRA + Medical) – PF. Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.

```
/*
Write a PL/SQL block which displays gross salary of employees as per user input
EID.
Consider EMP table with EID, EName, Deptno, Deptname, Gender, Age, BasicSal
with appropriate data types.
Gross_Salary:= BASICSAL + (DA + HRA + Medical) – PF.
Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.
Here for practical we have used Scott users inbuilt EMP table.
*/

SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF

DECLARE
   V_EID       EMP.EID%TYPE;
   V_NAME      EMP.ENAME%TYPE;
   BASIC_SAL   EMP.BASICSAL%TYPE;
   DA          NUMBER;
   HRA         NUMBER;
   MEDICAL     NUMBER := 500;
   PF          NUMBER;
   GROSS_SAL   NUMBER;

BEGIN
   V_EID := &EID;

   SELECT ENAME, BASICSAL
   INTO V_NAME, BASIC_SAL
   FROM EMP
   WHERE EID = V_EID;

   DA := BASIC_SAL * 0.50;
   HRA := BASIC_SAL * 0.15;
```

```
   PF := BASIC_SAL * 0.10;
   GROSS_SAL := BASIC_SAL + DA + HRA + MEDICAL - PF;

   DBMS_OUTPUT.PUT_LINE('--------------- SALARY SLIP ---------------');
   DBMS_OUTPUT.PUT_LINE('Employee ID     : ' || V_EID);
   DBMS_OUTPUT.PUT_LINE('Employee Name   : ' || V_NAME);
   DBMS_OUTPUT.PUT_LINE('Basic Salary    : Rs. ' || BASIC_SAL);
   DBMS_OUTPUT.PUT_LINE('DA Amount       : Rs. ' || DA);
   DBMS_OUTPUT.PUT_LINE('HRA Amount      : Rs. ' || HRA);
   DBMS_OUTPUT.PUT_LINE('Medical Amount  : Rs. ' || MEDICAL);
   DBMS_OUTPUT.PUT_LINE('Provident Fund  : Rs. ' || PF);
   DBMS_OUTPUT.PUT_LINE('--------------------------------------------');
   DBMS_OUTPUT.PUT_LINE('Gross Salary    : Rs. ' || GROSS_SAL);
   DBMS_OUTPUT.PUT_LINE('--------------------------------------------');
EXCEPTION
   WHEN NO_DATA_FOUND THEN
      DBMS_OUTPUT.PUT_LINE(
         'Employee record for given EID - ' || V_EID || ' does not exist'
      );
END;
/

SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p5.sql
Enter value for eid: 105
--------------- SALARY SLIP ---------------
Employee ID     : 105
Employee Name   : PRIYA
Basic Salary    : Rs. 32000
DA Amount       : Rs. 16000
HRA Amount      : Rs. 4800
Medical Amount  : Rs. 500
Provident Fund  : Rs. 3200
--------------------------------------------
Gross Salary    : Rs. 50100
--------------------------------------------
```

6. Write a PL/SQL block which accepts measurement in feet and displays it in cm, inch and meter.

```
/*
Write a PL/SQL block which accepts measurement in feet and displays it
in cm, inch and meter.

Formulas:-
        1)Inches :-feet * 12
        2)Centimeters :- feet * 30.48
        3)Meters:-feet*0.3045
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

DECLARE
   FEET   NUMBER;
   CM     NUMBER;
   INCH   NUMBER;
   METER  NUMBER;
BEGIN
   FEET := &FEET;

   CM := FEET * 30.48;
   INCH := FEET * 12;
   METER := FEET * 0.3048;

   DBMS_OUTPUT.PUT_LINE('Feet       : ' || FEET);
   DBMS_OUTPUT.PUT_LINE('Centimeter : ' || CM);
   DBMS_OUTPUT.PUT_LINE('Inch       : ' || INCH);
   DBMS_OUTPUT.PUT_LINE('Meter      : ' || METER);
END;
/

SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p5.sql
Enter value for eid: 105
--------------- SALARY SLIP ---------------
Employee ID      : 105
Employee Name    : PRIYA
Basic Salary     : Rs. 32000
DA Amount        : Rs. 16000
HRA Amount       : Rs. 4800
Medical Amount   : Rs. 500
Provident Fund   : Rs. 3200
--------------------------------------------
Gross Salary     : Rs. 50100
--------------------------------------------
```

7. Write a PL/SQL block which converts temperature from Celsius to Fahrenheit.

```
/*
Write a PL/SQL block which converts temperature from Celsius to
Fahrenheit.
Formulas:-
        -Celsius to Fahrenheit :-(Celsius *9/2)+32
        -Fahrenheit to Celsius :-(Fahrenheit-32)/5/9
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF
ACCEPT CH PROMPT 'Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to
Celsius: '
DECLARE
         CELSIUS NUMBER(10,2);
        FAHRENHEIT NUMBER(10,2);
         CHOICE NUMBER(10);
BEGIN
   CHOICE := &CH;
   IF CHOICE = 1 THEN
      CELSIUS := &CELSIUS;
      FAHRENHEIT := (CELSIUS * 9 / 5) + 32;

     DBMS_OUTPUT.PUT_LINE('Value of FAHRENHEIT :- ' || FAHRENHEIT);

   ELSIF CHOICE = 2 THEN
           FAHRENHEIT := &FAHRENHEIT;
           CELSIUS := (FAHRENHEIT - 32) * 5 / 9;
      DBMS_OUTPUT.PUT_LINE('Value of CELSIUS :- ' || CELSIUS ||'C');
   ELSE
      DBMS_OUTPUT.PUT_LINE('INVALID CHOICE... Enter 1 or 2');
   END IF;
END;
/
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p7.sql
Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to Celsius: 1
Enter value for celsius: 12
Enter value for fahrenheit: 30
Value of FAHRENHEIT :- 53.6
```

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p7.sql
Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to Celsius: 2
Enter value for celsius: -45
Enter value for fahrenheit: 45
Value of CELSIUS :- 7.22C
SQL>
```

8. Write a PL/SQL block to delete the record of employee for given EID.

```
/*
Write a PL/SQL block to remove an employee record
for the given Employee ID.
*/

SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF
DECLARE
   EMP_ID EMP.EID%TYPE;
   CNT NUMBER;
BEGIN
   EMP_ID := &EID;

   SELECT COUNT(*) INTO CNT FROM EMP WHERE EID = EMP_ID;

   IF CNT > 0 THEN
      DELETE FROM EMP WHERE EID = EMP_ID;

      COMMIT;
      DBMS_OUTPUT.PUT_LINE( 'Employee ID ' || EMP_ID || ' deleted successfully.'
);
   ELSE
      DBMS_OUTPUT.PUT_LINE('Employee ID ' || EMP_ID || ' does not exist.' );
   END IF;

END;
/
SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p8.sql
Enter value for eid: 105
Employee ID 105 deleted successfully.
SQL>
```

9. Write a PL/SQL block that calculates the simple interest based on given principal amount, rate of interest and number of years.

```
/* Write a PL/SQL block to calculate Simple Interestand Total Amount. */

SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF
DECLARE
   P NUMBER;
   R NUMBER;
   T NUMBER;
   SI NUMBER;
   AMOUNT NUMBER;
BEGIN
   P := &P;
   R := &R;
   T := &T;
   SI := (P * R * T) / 100;
   AMOUNT := P + SI;
  DBMS_OUTPUT.PUT_LINE('=======================');
   DBMS_OUTPUT.PUT_LINE('Principal Amount : ' || P);
   DBMS_OUTPUT.PUT_LINE('Rate of Interest : ' || R || '%');
   DBMS_OUTPUT.PUT_LINE('Number of Years  : ' || T);
   DBMS_OUTPUT.PUT_LINE('Simple Interest  : ' || SI);
   DBMS_OUTPUT.PUT_LINE('Total Amount     : ' || AMOUNT);
END;
/
SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p9.sql
Enter value for p: 12000
Enter value for r: 10
Enter value for t: 5
=======================
Principal Amount : 12000
Rate of Interest : 10%
Number of Years  : 5
Simple Interest  : 6000
Total Amount     : 18000
```

10. Write a PL/SQL block to calculate the square and cube of the given number.

```
/*
Write a PL/SQL block to calculate the square and cube
of the given number.
*/

SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF

DECLARE
   N NUMBER;
   SQUARE NUMBER;
   CUBE NUMBER;
BEGIN
   N := &N;
   SQUARE := N * N;
   CUBE := N * N * N;

   DBMS_OUTPUT.PUT_LINE('Given Number : ' || N);
   DBMS_OUTPUT.PUT_LINE('Square       : ' || SQUARE);
   DBMS_OUTPUT.PUT_LINE('Cube         : ' || CUBE);

END;
/

SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p10.sql
Enter value for n: 20
Given Number : 20
Square       : 400
Cube         : 8000
SQL>
```

11. Write a simple procedure that increases by the salary of employees for the given department no. by percentage inputted by the user using IN parameter. Also handle the exception if inputted department number not found.

```
/*
Write a simple procedure that increases by the salary of employees for the given
department no. by percentage inputted by the user using IN parameter. Also
handle the exception if inputted department number not found.
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

CREATE OR REPLACE PROCEDURE increase_salary (
   p_deptno  IN NUMBER,
   p_percent IN NUMBER
)
IS
   v_count NUMBER;
BEGIN
   SELECT COUNT(*)
   INTO v_count
   FROM EMP
   WHERE DEPTNO = p_deptno;

   IF v_count = 0 THEN
      RAISE_APPLICATION_ERROR(
         -20001,
         'Department number ' || p_deptno || ' not found'
      );
   END IF;

   UPDATE EMP
   SET BASICSAL = BASICSAL + (BASICSAL * p_percent / 100)
   WHERE DEPTNO = p_deptno;

   COMMIT;

   DBMS_OUTPUT.PUT_LINE(
```

```
    'Salary increased by ' || p_percent ||
    '% for department ' || p_deptno
  );

EXCEPTION
  WHEN OTHERS THEN
    DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
END;
/
SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\92600584022\DBMS\Unit_2\p11.sql
SQL> VARIABLE v_deptno NUMBER;
SQL> VARIABLE v_percent NUMBER;
SQL> EXEC :v_deptno := 10;
SQL> increase_salary(:v_deptno, :v_percent);
SQL> select * from emp;

       EID ENAME                     DEPTNO
---------- ---------------------- ------
DEPTNAME                  GENDER        AGE   BASICSAL
---------- ------------ --------- ------- ---------
       101 KUNJAL                        10
IT                        MALE          21  113924.95

       104 AMAN                          30
SALES                     MALE          25   28000
```

12. Write a procedure that search whether the given employee number is present or not in the table if present then print name, salary and department number of that employee. (Use both IN and OUT mode variables) and also Write a PL/SQL block to call the SEARCH_EMP procedure.

```
/*
Write a procedure that search whether the given employee number is
present or not in the table if present then print name, salary and
department number of that employee. (Use both IN and OUT mode
variables) and also Write a PL/SQL block to call the SEARCH_EMP
procedure.
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

CREATE OR REPLACE PROCEDURE SEARCH_EMP (
   P_EID       IN NUMBER,
   P_ENAME     OUT VARCHAR2,
   P_SALARY    OUT NUMBER,
   P_DEPTNO    OUT NUMBER
)
IS
BEGIN
   SELECT ENAME, BASICSAL, DEPTNO
   INTO P_ENAME, P_SALARY, P_DEPTNO
   FROM EMP
   WHERE EID = P_EID;

EXCEPTION
   WHEN NO_DATA_FOUND THEN
      P_ENAME := NULL;
      P_SALARY := NULL;
      P_DEPTNO := NULL;

      DBMS_OUTPUT.PUT_LINE(
         'Employee number ' || P_EID || ' not found.'
      );
END;
/
```

SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON

Output:

```
SQL> VARIABLE v_eid NUMBER;
SQL> VARIABLE v_ename VARCHAR2(30);
SQL> VARIABLE v_salary NUMBER;
SQL> VARIABLE v_deptno NUMBER;
SQL> EXEC :v_eid := 104;
SQL> EXEC SEARCH_EMP(:v_eid, :v_ename, :v_salary, :v_deptno);
SQL> @ D:\92600584022\DBMS\Unit_2\p12_plsql.sql
Employee Name : KUNJAL
Salary        : 113924.95
Department No : 10
```

13. Write a procedure that accept employee ID and remove records from employee table. Also create a calling program to call procedure

```
/*
Write a procedure that accept employee ID and remove records from employee
table. Also create a calling program to call procedure.
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

CREATE OR REPLACE PROCEDURE REMOVE_EMPLOYEE (
   P_EID IN EMP.EID%TYPE
)
IS
BEGIN
   DELETE FROM EMP
   WHERE EID = P_EID;

   IF SQL%ROWCOUNT = 0 THEN
      DBMS_OUTPUT.PUT_LINE(
         'Employee ID ' || P_EID || ' not found.'
      );
   ELSE
      DBMS_OUTPUT.PUT_LINE(
         'Employee ID ' || P_EID || ' deleted successfully.'
      );
   END IF;

END;
/

SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\92600584022\DBMS\Unit_2\p13.sql
SQL> SET SERVEROUTPUT ON;
SQL> VARIABLE v_eid NUMBER;
SQL> EXEC :v_eid := 102;
SQL> EXEC REMOVE_EMPLOYEE(:v_eid);
Employee ID 102 deleted successfully.
SQL>
```

14. Write a function that returns the age, category of an employee.

Categories:

- **Age < 18 → Minor**
- **18–60 → Adult**
- **60 → Senior**

```
/*
Write a function that returns the age, category of an employee.
Categories:
● Age < 18 → Minor
● 18–60 → Adult
● 60 → Senior
*/
SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

CREATE OR REPLACE FUNCTION GET_AGE_CATEGORY (
   P_EID IN NUMBER
)
RETURN VARCHAR2
IS
   V_AGE EMP.AGE%TYPE;
   V_CATEGORY VARCHAR2(20);
BEGIN
   SELECT AGE
   INTO V_AGE
   FROM EMP
   WHERE EID = P_EID;

   IF V_AGE < 18 THEN
      V_CATEGORY := 'Minor';
   ELSIF V_AGE <= 60 THEN
      V_CATEGORY := 'Adult';
   ELSE
      V_CATEGORY := 'Senior';
   END IF;

   RETURN 'Age: ' || V_AGE || ', Category: ' || V_CATEGORY;
```

```
EXCEPTION
   WHEN NO_DATA_FOUND THEN
      RETURN 'Employee not found';
END;
/

SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> VARIABLE v_eid NUMBER;
SQL> VARIABLE v_result VARCHAR2(100);
SQL> EXEC :v_eid := 103;

SQL> EXEC GET_AGE_CATEGORY(:v_eid);

V_RESULT
----------------- ---------------------------------------
Name: PRIYA,Age: 23, Category: Adult
```

15. Write a PACKAGE which includes Package Declaration, Package Body and call the individual object from the Package.

```
/*
Write a PACKAGE which includes Package Declaration, Package Body and call the
individual object from the Package.
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

CREATE OR REPLACE PACKAGE my_package
AS
   PROCEDURE greet;
END my_package;
/

CREATE OR REPLACE PACKAGE BODY my_package
AS
   PROCEDURE greet
   IS
   BEGIN
      DBMS_OUTPUT.PUT_LINE('Hello World');
   END greet;
END my_package;
/

BEGIN
   my_package.greet;
END;
/
SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @D:\92600584022\DBMS\Unit_2\p15.sql
Package successfullly created..
```